and atoms, molecules, crystals, 166, 169, 372
and atomic nuclei, 169, 184, 199; *see also* Fusion; Nuclear force; Nucleus, atomic
and superconductivity, 231, 339
and superfluidity, 186–87, 186n, 208
and spectral lines of light, 335
and entropy, 423, 445–46
and quantum hair on black holes, 284
and time machines, 515, 517–23
on macroscopic scales, 372–76,
marriage with special relativity, 150–52, 160
marriage with general relativity, *see* Quantum gravity
speculations about failure of, 184, 207
see also other Quantum entries: Degeneracy; Hawking radiation; Probability; Uncertainty principle; Vacuum fluctuations; Wave-particle duality
Quantum nondemolition, 375–76, 377f, 386–87
Quarks, search for, 370–71
Quasars:
overview of, 45–48, 47f
discovery of, 335–37
variability of, 337–38
energy of, 339
speculations about power source, 273, 339–41
gigantic black holes as power source, 346–54, 407–10, 408f
role of accretion disks in, 346–54
detailed model of, 351–53, 352f

Radio galaxies:
discovery of, 330–31
discovery of double radio lobes, 333
further observations of, 334
energy requirements of, 339
speculations about power source, 273, 339–41
gigantic black holes as power source, 346–54, 407–10, 408f
role of accretion disks in, 346–54
detailed model of, 351–53, 352f
see also Cygnus A; Quasars; Radio jets
Radio jets:
discovery of, 343–45
gyroscopic stability of source, 345
gigantic black hole as source, 345–54, 407–10, 408f
Radio telescopes:
Jansky's and Reber's, 323–24, 234f–25f
relation to radar, 327
interferometers, principle of, 328–30, 329f
Jodrell Bank, 331–33, 332f–33f
Greenbank and Owens Valley, 334
Very Large Array (VLA), 343, 344f, 345